Homework 5

ECON 470, Spring 2026

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1. Mean Uncompensated Care by Year

Table 1: Mean hospital uncompensated care ($mm) by year, 2011–2018

Year	Mean	SD	N
2011	6.240000	18.420000	2920
2012	7.760000	35.680000	4924
2013	8.250000	52.930000	4785
2014	7.110000	24.100000	4780
2015	5.980000	18.580000	4747
2016	8.460000	99.520000	4733
2017	9.150000	33.480000	4441
2018	8.810000	22.720000	4379

Q1 answer: Uncompensated care rose from 6.24mm per hospital in 2011 to 8.25mm in 2013. It then fell to 7.11mm in 2014 and further to 5.98mm in 2015, which is consistent with the ACA's Medicaid expansion taking effect. Newly insured patients generated reimbursable claims rather than uncompensated care. However, the trend does not sustain a clean decline. The mean rose sharply in 2016 and peaked at 9.15mm in 2017. The 2016 figure carries an unusually large standard deviation of 99.52mm, indicating a small number of hospitals with extreme values are pulling the mean up rather than a genuine system-wide reversal.

2. Uncompensated Care Over Time by Expansion Status

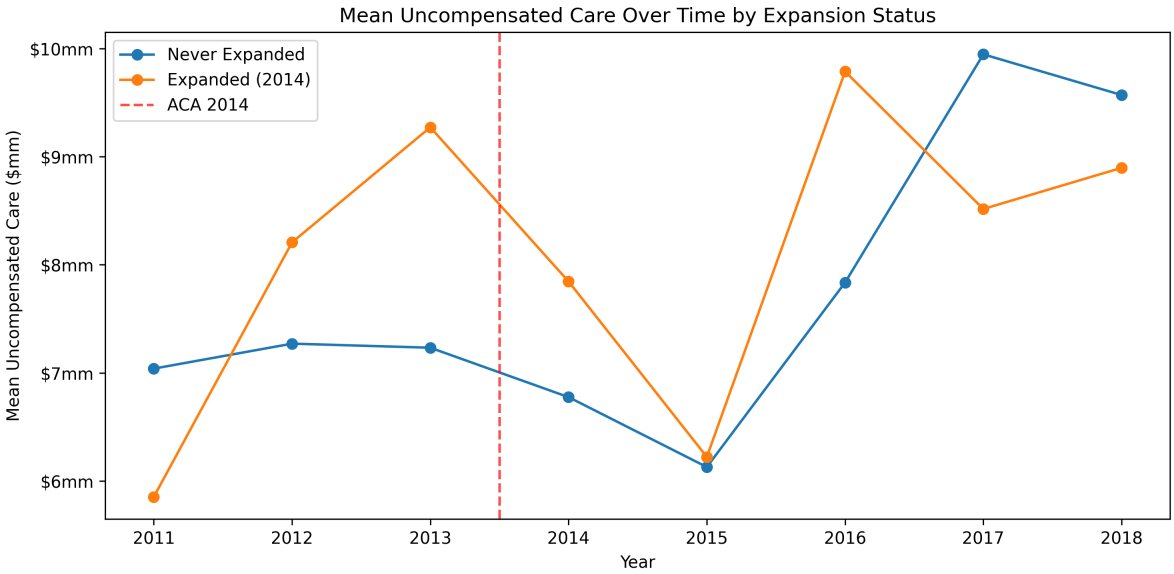

Figure 1: Mean uncompensated care over time for 2014 expanders vs never-expanders

Q2 answer: The graph does suggest a treatment effect. Before 2014 the two groups tracked each other fairly closely — expanders averaged 9.27mm and never-expanders 7.23mm in 2013. After expansion, both groups declined, but expanding states fell more sharply, reaching 6.22mm by 2015 compared to 6.13mm for never-expanders. The gap opening up right at 2014 is consistent with Medicaid expansion converting previously uninsured patients into covered ones, reducing the care hospitals had to absorb for free.

3. 2x2 Difference-in-Differences Table

Table 2: DD Estimate: -0.84mm

Group	Pre (2012)	Post (2015)	Difference
Never Expanded	7.270000	6.130000	-1.140000
Expanded (2014)	8.210000	6.220000	-1.980000

4. Policy Discussion

Q4 answer: Several forces were at work during this period. The most direct is Medicaid expansion itself which covered adults up to 138% of the federal poverty level. This means that hospitals could bill for care they previously wrote off entirely. But the ACA's insurance exchanges were also pulling in uninsured people in non-expansion states, which partly explains why never-expanders also saw declines. The broader economic recovery from the 2008 recession reduced the share of patients unable to pay their bills. Also, S-10 reporting standards shifted around 2012-2013, so some of the measured change reflects accounting practices rather than actual care delivered. Taken together, we expect expansion to reduce uncompensated care specifically because insured patients generate reimbursable claims. Hospitals get paid instead of absorbing the cost.

5. Standard DD Regression

Table 3: DD regression estimates of Medicaid expansion on uncompensated care ($mm)

	Coef	SE
Intercept	7.200	(1.105)
Post (year $\geq$ 2014)	0.809	(1.370)
Expanded	0.881	(1.289)
Treat (Post $\times$ Expanded)	-0.649	(1.599)
N	25,391	
R^2	0.000	

6. TWFE with Hospital and Year Fixed Effects

Table 4: TWFE estimates: hospital and year fixed effects, state-clustered SEs (2014 expanders and never-expanders)

	Estimate
Treat	-1.268
	(1.884)
N	25,197

Q6 answer: Adding hospital and year fixed effects changes the estimate noticeably. The simple DD (Q5) gave -0.649mm (SE 1.599, p=0.685), while the TWFE estimate is -1.268mm (SE 1.884, p=0.505) on 25,197 observations. Hospital fixed effects absorb permanent differences across hospitals — size, teaching status, location — that were biasing the simple DD upward. Year fixed effects remove common trends affecting all hospitals regardless of expansion status. Neither estimate is statistically significant, but both are negative and point in the expected direction. The TWFE estimate is larger in magnitude than the simple DD, suggesting OLS was understating the effect of expansion.

7. TWFE with All States

Table 5: TWFE estimates: all states, hospital and year fixed effects, state-clustered SEs

	Estimate
Treat	-0.643
	(1.321)
N	35,462

Q7 answer: Including all states shifts the estimate from -1.268mm to -0.643mm, moving it closer to zero. This reflects a known problem with TWFE under staggered adoption — when states expand at different times, already-treated states get used as implicit controls for later-treated ones. If treatment effects differ across expansion cohorts or evolve over time, this contaminates the estimate. The fact that the coefficient shifts when late-expanders are added is a signal that those dynamics are present here.

8. Event Study: 2014 Expanders vs Never-Expanders

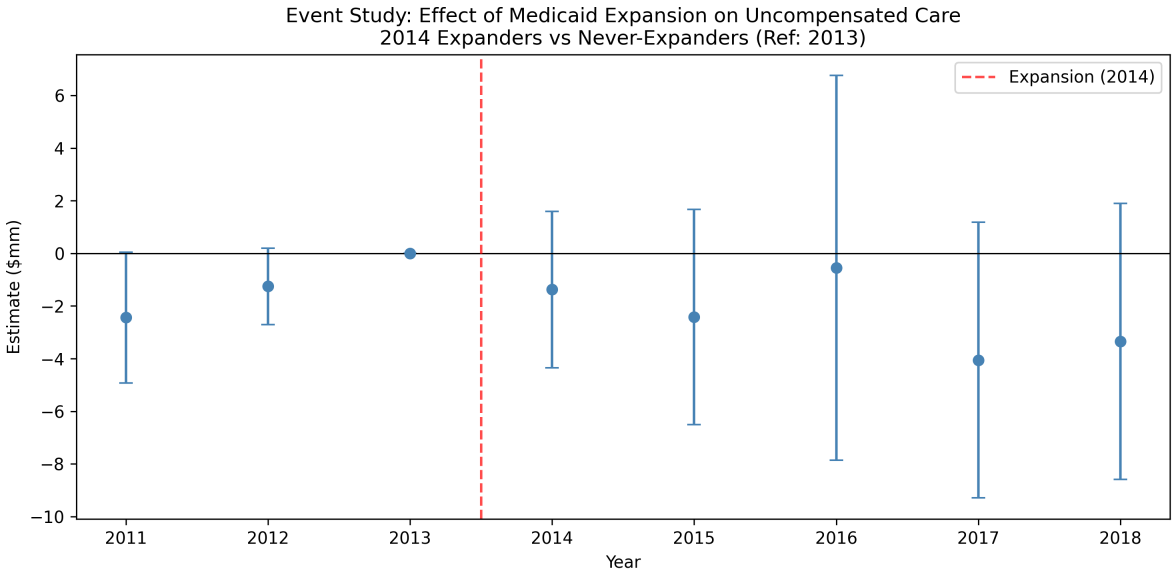

Figure 2: Event study: effect of Medicaid expansion on uncompensated care, 2014 expanders vs never-expanders (ref: 2013)

9. Event Study: All States (Event Time)

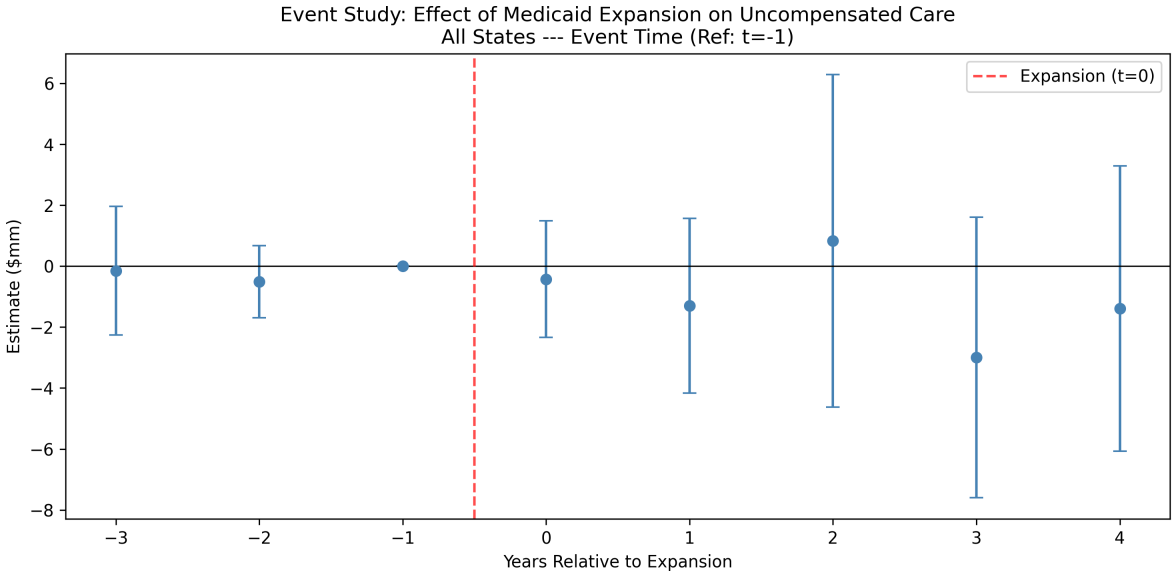

Figure 3: Event study: effect of Medicaid expansion on uncompensated care, all states on event time (ref: t=-1)

10. Summary and TWFE Limitations

Q10 answer: Across Questions 5 through 9, the results consistently point toward Medicaid expansion reducing uncompensated care, though none of the estimates reach statistical significance. The simple DD (Q5) gave -0.649mm, TWFE on the 2014 sample (Q6) gave -1.268mm, and adding all states (Q7) produced -0.643mm — all negative and in the expected direction.

The event study in Q8 raises some concern about parallel trends. The pre-period coefficients for 2011 (-2.440mm, SE 1.266) and 2012 (-1.250mm) are not close to zero, suggesting expanding and non-expanding states may have been on somewhat different trajectories before 2014. If that pre-existing divergence is real, our DD estimates partly capture it rather than the effect of expansion alone. Post-expansion, the coefficient for 2015 is -2.414mm and remains negative through the end of the sample, consistent with a persistent reduction in uncompensated care, though the wide confidence intervals make it difficult to pin down the magnitude with precision.

The core limitation of TWFE with staggered adoption is that it uses already-treated states as controls for later-treated ones. If early and late expanders experienced different effect sizes, those comparisons inject bias and can even produce negative weights on some underlying 2x2 DDs. There are other alternatives that can estimate cohort-specific effects before aggregating, which avoids this problem by never using treated units as controls.